

Introduction & Hook

Ask:

“If you build a VR app today, how will it run on different headsets like Quest, Vive, or HoloLens?”

Explain:

Without standards, developers would need to rewrite apps for each device. This is why **frameworks and SDKs** exist.

VR Development Ecosystem

What are Frameworks and SDKs?

◆ Framework

A framework provides a structure and environment to build applications.

Examples:

- Unity XR framework
- Unreal Engine

Framework includes:

- Rendering system
 - Physics engine
 - Scene management
-

◆ SDK (Software Development Kit)

An SDK provides tools, libraries, and APIs to interact with specific hardware.

Examples:

- Meta XR SDK
- SteamVR SDK
- Windows Mixed Reality SDK

SDK handles:

- Device input
- Tracking data
- Controller interaction

♦ Simple Analogy

- Framework = Building structure
 - SDK = Tools to connect devices
-

Relationship Between Components

Explain pipeline:

```
Application
↓
Framework (Unity/Unreal)
↓
SDK (OpenXR / SteamVR)
↓
Drivers
↓
Hardware (Headset)
```

This is very important for exams.

Popular VR Frameworks & SDKs

Common Tools

Frameworks:

- Unity (most popular for beginners)
- Unreal Engine (high-end graphics)

SDKs:

- Meta XR SDK
- SteamVR SDK
- Windows MR SDK
- OpenXR (standard SDK)

Explain:

Framework builds the app, SDK connects it to hardware.

OpenVR

What is OpenVR?

Definition:

OpenVR is an API developed by Valve that allows VR applications to interact with VR hardware through SteamVR.

Used with:

- HTC Vive
 - Valve Index
-

Key Features:

- Device-independent API
 - Supports multiple headsets
 - Works with SteamVR ecosystem
-

Limitations:

- Not fully standardized across industry
 - Tied to Steam ecosystem
 - Being replaced by OpenXR
-

OpenXR (Modern Standard)

What is OpenXR?

Definition:

OpenXR is an open, royalty-free standard developed by the Khronos Group for building cross-platform VR/AR applications.

Why OpenXR is Important

Before OpenXR:

- Different SDK for each device
- Code duplication

After OpenXR:

- One API → multiple devices
-

Key Advantages:

- Cross-platform compatibility
- Device-independent development
- Industry standard (future-proof)
- Supported by:
 - Meta (Quest)
 - Microsoft (HoloLens)
 - HTC
 - Valve

Comparison: OpenVR vs OpenXR

Feature	OpenVR	OpenXR
Developer	Valve	Khronos Group
Platform	SteamVR	Cross-platform
Standardization	Limited	Industry standard
Future relevance	Declining	Growing

Important Teaching Line:

OpenXR is the future of XR development.

Real Development Example

Explain:

In Unity VR app:

1. Developer builds scene
 2. Unity uses OpenXR plugin
 3. OpenXR communicates with headset
 4. App runs on multiple devices
-

Recap & Questions

Key Points:

- Framework = development environment
 - SDK = hardware interface tools
 - OpenVR = older API (Steam-based)
 - OpenXR = modern cross-platform standard
-

Quick Questions:

1. What is difference between framework and SDK?
2. Why is OpenXR important?
3. What problem does OpenXR solve?
4. Which API is replacing OpenVR?